

AB POWER ENGINEERING

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1

ENGINE FUEL AND FUEL SYSTEM

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2

1. Conventional fuel used in internal combustion engine.
 - a. Producer gas
 - b. Biogas
 - c. Bioethanol
 - d. Gasoline
 - e. None of the above
2. Which of the following fuels can be used fully for compressed-ignition engine?
 - a. Producer gas
 - b. Biogas
 - c. Bioethanol
 - d. Gasoline
 - e. None of the above
3. Which of these properties determine the quality of fuel?
 - a. Flash point
 - b. Volatility
 - c. Calorific value
 - d. Ignition quality
 - e. All of the above
4. Property of fuel which is indicated by the lowest temperature where its vapor starts to ignite.
 - a. Flash point
 - b. Volatility
 - c. Calorific value
 - d. Ignition quality
 - e. None of the above
5. Refers to the ignition quality of diesel fuel.
 - a. Octane Number
 - b. Cetane Number
 - c. All of the above
 - d. None of the above
6. Violent noises heard in engine giving pinging sound during the process of combustion.
 - a. Detonation
 - b. Vibration
 - c. Explosion
 - d. None of the above

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3

7. Which of the following is not a function of carburetor?
 - a. To mix air and fuel thoroughly.
 - b. To regulate the air-fuel ratio at different speed and loads.
 - c. To supply the correct amount of fuel.
 - d. None of the above
8. A device that is recently used in diesel engine to preheat diesel fuel for ease of start up.
 - a. Atomizer
 - b. Glow plug
 - c. Spark plug
 - d. None of the above
9. Which is not a function of fuel- injection pump in diesel engines?
 - a. Measure the correct amount of fuel.
 - b. Maintain the correct timing for starting to the ending of injection.
 - c. Inject the fuel into the combustion chamber.
 - d. Atomize the fuel for quick ignition.
 - e. None of the above
10. If carburetor is for gasoline, fuel- injection pump and spray nozzle is for _____.
 - a. kerosene
 - b. naphtha
 - c. diesel
 - d. All of the above
11. Device provided for gasoline-fueled engines to deliver the fuel from the fuel- injection pump of the engine cylinder.
 - a. Transfer pump
 - b. Lift pump
 - c. Fuel-injection pump
 - d. All of the above

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4

12. Governor hunting in engine refers to when the engine experience an erratic variation in the engine ____ that is too fast and then too slow.

- a. power
- b. torque
- c. speed
- d. None of the above

13. Mechanical device that controls the engine to a nearly constant speed under different load conditions.

- a. Carburetor
- b. Fuel pump
- c. Governor
- d. None of the above

14. Tractors are always fitted with ____ that controls the supply of fuel to the engine.

- a. governor
- b. accelerator pedal
- c. clutch
- d. None of the above

15. An attachment into the fuel-supply system of the engine that removes solid particles and dirt.

- a. Oil filter
- b. Air filter
- c. Fuel filter
- d. All of the above

16. Octane number of gasoline fuel.

- a. 92 – 98
- b. -25
- c. 107
- d. None of the above

17. Cetane number of diesel fuel.

- a. 5
- b. 45 – 55
- c. -2
- d. None of the above

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5

18. Which of the following is an effect of detonation in the engine?

- a. Rise in pressure.
- b. Loss of power.
- c. overheating of engine.
- d. All of the above

19. Problems encountered in fuel system of an engine.

- a. Clogged fuel filters
- b. Dirty fuel injector
- c. Leaking fuel lines
- d. Malfunctioning of fuel pressure regulators
- e. All of the above

20. Which of the following is a symptom of failing fuel system?

- a. Reduce engine power
- b. Difficulty of starting the engine
- c. Poor acceleration
- d. Engine misfiring
- e. All of the above

21. Common symptoms of fuel system problem.

- a. Black smoke from the exhaust
- b. Poor fuel economy
- c. Engine reduce performance
- d. All of the above

22. Things to consider when trouble shooting fuel system.

- a. Check fuel tank if there is fuel
- b. Check possibilities of contaminants
- c. Check injection pump
- d. Check fuel pump if working
- e. Check if fuel and air filter are blocked
- f. Check for the possibilities of air or fuel leak
- g. All of the above

23. Which of the following is not included in the components of fuel system?

- a. Fuel tank
- b. Fuel lines
- c. Fuel pump
- d. Fuel filter
- e. Carburetor
- f. Fuel injector
- g. Fuel pressure regulator
- h. None of the above

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6

24. The mass of fuel consumed per unit of power produced, typically expressed in grams per kilowatt-hour (g/kWh) or pounds per horsepower-hour (lb/hp·h).

- a. Fuel consumption rate
- ☒ b. Specific fuel consumption
- c. Fuel consumed per load
- d. None of the above

25. Another term used for specific fuel consumption?

- ☒ a. Brake specific fuel consumption
- b. Fuel consumption rate
- c. Rate of consumption of fuel
- d. None of the above

26. Specific fuel consumption of gasoline engine.

- ☒ a. 0.20 to 0.50 kg/kWh
- b. 0.12 to 0.22 kg/kWh
- c. 0.09 to 0.19 kg/kWh
- d. None of the above

27. Specific fuel consumption of diesel engine.

- a. 0.20 to 0.50 kg/kWh
- ☒ b. 0.12 to 0.22 kg/kWh
- c. 0.09 to 0.19 kg/kWh
- d. None of the above

28. Factors affecting the actual values of specific fuel consumption.

- a. Engine Type
- b. Load
- c. Operating Conditions
- ☒ d. All of the above

29. Which of the following factors influences specific fuel consumption?

- a. Engine load
- b. Engine speed
- c. Engine design
- d. Fuel type
- ☒ e. All of the above

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7

30. Lower SFC values indicate ____ fuel efficiency, with consumption increasing at lower power loads, especially at idle.

- a. lesser
- b. greater
- c. the same
- d. None of the above

31. Understanding SFC helps in identifying _____ operating points for an engine to minimize fuel consumption.

- a. Optimal
- b. Maximum
- c. Minimum
- d. All of the above

32. A 10 hp engine is use in operating a rice thresher. What is amount of fuel it may consumed in 1 hour if its SFC is 0.20 kg/kwh? Assume a density of 0.7 kg/liter.

- a. 2.13 liters per hour
- b. 3.41 liters per hour
- c. 4.05 liters per hour
- d. None of the above

33. A 16 hp engine is use in operating a bio-fertilizer pulverizing machine. What is amount of fuel it may consumed in 1 hour if its SFC is 0.22 kg/kwh?

Assume a density of 0.9 kg/liter.

- a. 2.92 liters per hour
- b. 3.45 liters per hour
- c. 4.12 liters per hour
- d. None of the above

$$0.2 \text{ kg/kW-hr} \times 10 \text{ hp} \times 0.764 \text{ kW/hp} / 0.7 \text{ kg/liter} = 2.13 \text{ liters per hour}$$

$$0.22 \text{ kg/kW-hr} \times 16 \text{ hp} \times 0.746 \text{ kW/hp} / 0.9 \text{ kg/liter} = 2.92 \text{ liters per hour}$$

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8

GOD BLESS !!!

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